

English

Work
Book
4~6 age

Computational Thinking

Detective

Dalton & Terru¹

"Missing Cat Tom"

Teacher's Guide
Free Download

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Children dream and have ability a dream come true.

Crevers is a new method of coding education brand that can solve problems with computational thinking through storytelling.

Create Your Universe!



**Computational thinking Detective Dalton & Terry1
Missing cat Tom
By Crevers**

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Crevers wishes that children all over the world can
get same computational thinking education
regardless of economic power.

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opportunities of education and
experience to make children dream and
cultivate people of talent.

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can reach opportunity of learning.





01

What is computational thinking?

Computational thinking is thinking that solves problem effectively by using scientific conceptual principle of computer.

If you want to solve a problem with a computer, you need to simplify a complex problem and give logical and efficient orders in sequence to make computer solve.

At this moment, computational thinking is to think solutions logically and sequentially by recognizing a problem and giving shape to creative solutions.

Computational thinking **increases ability to solve problems** in real life and various areas.

02

5 Elements of Computational Thinking Focused Learning in Crevers.

Algorithm

It is procedural expression of action or ways to solve a problem.

Sequence

It means order of sequential events or actions.

Loop

It means repeating action or method the number of times.

Decomposition

It means dividing the problem into smaller units.

Branching

A condition that determines options.
For example, if the weather is sunny, you can go to playground, but if the weather is rainy, you cannot go.

Composition of this book



Contents	
Chapter 1 Missing Cat Tom • week 1 Objectives: Understand the concept of missing information and how to find it. Activities: Find the missing cat in the picture. Example: Find the missing cat in the picture.	Chapter 2 Find Cat Tom • week 2 Objectives: Understand the concept of finding information and how to find it. Activities: Find the missing cat in the picture. Example: Find the missing cat in the picture.
Chapter 3 Who is the criminal? • week 3 Objectives: Understand the concept of identifying the criminal and how to find it. Activities: Find the missing cat in the picture. Example: Find the missing cat in the picture.	Chapter 4 Chase the Criminal • week 4 Objectives: Understand the concept of chasing the criminal and how to find it. Activities: Find the missing cat in the picture. Example: Find the missing cat in the picture.

Contents

You can check the principle of computational thinking of each chapter.

Guide	
Chapter 1 Missing Cat Tom p.10 -11 Objectives: Find out cat's characteristics such as eyes, nose, fur, and etc. and figure out how to collect data. Activities: Find out cat's characteristics such as eyes, nose, fur, and etc. and figure out how to collect data. Example: Find out cat's characteristics such as eyes, nose, fur, and etc. and figure out how to collect data.	Chapter 2 Missing Cat Tom p.12 -13 Objectives: Learn how to figure out characteristics of cat and compare cats in the pictures. Activities: Learn how to figure out characteristics of cat and compare cats in the pictures. Example: Learn how to figure out characteristics of cat and compare cats in the pictures.

Teacher's Guide

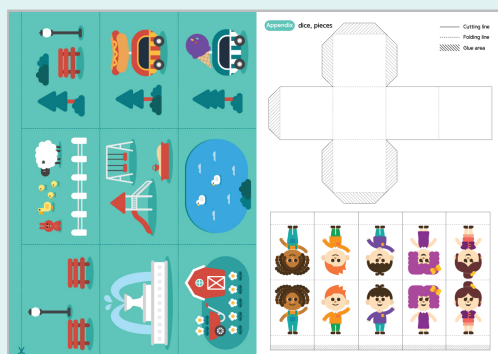
Teacher's Guide(PDF) can be downloaded at www.crevers.com

It contains how to guide the Workbook effectively.



Workbook

It is workbook that children can enjoy and learn computational thinking naturally.



Appendix

We added fun to the Workbook by cutting and pasting activities.

Contents



Chapter 1

Missing Cat Tom • week 1

Recognize problem

Grasp a situation and information of voice by giving restriction to accept case of the lost cat over the phone with client.

Branching, Abstraction

Abstract information by finding out characteristics of cat while drawing cat montage based on voice information.

Decomposition, Branching

Learn **decomposition analyzing** characteristics of cats and solve problems by gathering information by **matching** with missing cat, assuming that cats were caught partially on security camera in the park.

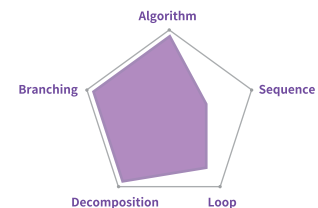
Loop

You need to repeat looking at partial characteristics of the cat and comparing with the missing cat. You can find out about '**Loop**' by repeating this as much the number of cat pictures taken on security camera.

After Class



- While reading a book with your child, draw other cats and play cat search game taken on camera together.
- Please give process-centered praise and encouragement for child.



Chapter 2

Find Cat Tom • week 2

Data collection

Find out about method how to **collect data** with voice information by giving restriction talking over the phone with a witness.

Branching, Abstraction

While completing the park based on voice information, learn how to locate with what criteria and learn the process of **abstracting** information that gives shape to the information you need.

Decomposition, Branching

Learn **decomposition analyzing** characteristics of cats and solve problems by gathering information and **matching** with the missing cat, assuming that cats are hiding in the park.

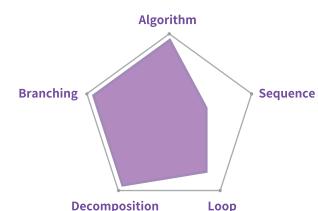
Loop

You need to repeat looking at partial characteristics of the cat and comparing with the missing cat. You can find out about '**Loop**' by repeating this as much the number of cats.

After Class



- While reading a book with your child, talk about coordinate of the park for right and left and play cat search game together.
- Please give process-centered praise and encouragement for child.





Chapter 3

Who is the criminal? • week 3

Data Representation

Find out how to **express information** by identifying criminal's appearance and drawing clothes with voice by giving restriction talking over the phone with an witness.

Algorithm, Sequence

By looking at the map, you can learn about 'Algorithm' and 'Sequence' in the process of finding out sequence of criminal's route tracking based on a witness' testimony.

Decomposition, Branching

While observing Criminal's description closely and grasping characteristics of the criminal, learn how to 'Decomposition' in the process of finding and **matching** suspects.

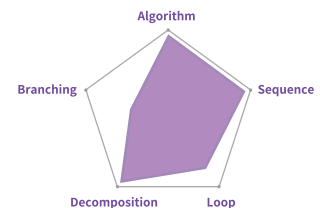
Loop

You can find out about 'Loop' by knowing partial characteristics of the criminal, comparing with the suspects, and **repeating** same process.

After Class



- While reading a book with your child, give several hints to make other routes and play criminal search game together.
- Please give process-centered praise and encouragement for child.



Chapter 4

Chase the Criminal • week 4

Problem Abstraction

Find out **necessary elements to solve problem** by looking and combining remaining evidence in site and testimony of a witness.

Decomposition, Branching

Learn about 'Branching' and 'Decomposition' in the process of figuring out where the destination is with what kind of rules according to the color and pattern of the bus by looking at bus tickets.

Sequence, Loop

Learn about 'Sequence' by passing several areas in order to find the criminal and figure out about 'Loop' in the process of repeating game with dice.

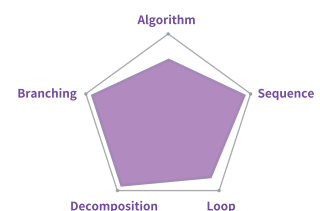
Branching

Mark red color for even numbers and blue colors for odd numbers. Depending on number of dice, move your horse by dividing even and odd numbers and learn about **condition**.

After Class



- While reading a book with your child, make various buses by referring bus tickets, and play criminal tracing game together.
- Please give process-centered praise and encouragement for child.



Hi, I am Terry.
We are detectives.
We opened detective office today.

Hi, I am Dalton.
Let's solve the case.

Detective Office



Ring! Ring!

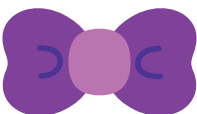
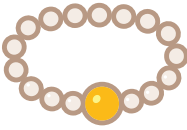
Yes, it's Dalton and Terry's
detective office.

Our cat Tom is missing.
Please find!



Case book

MONTAGE GAME

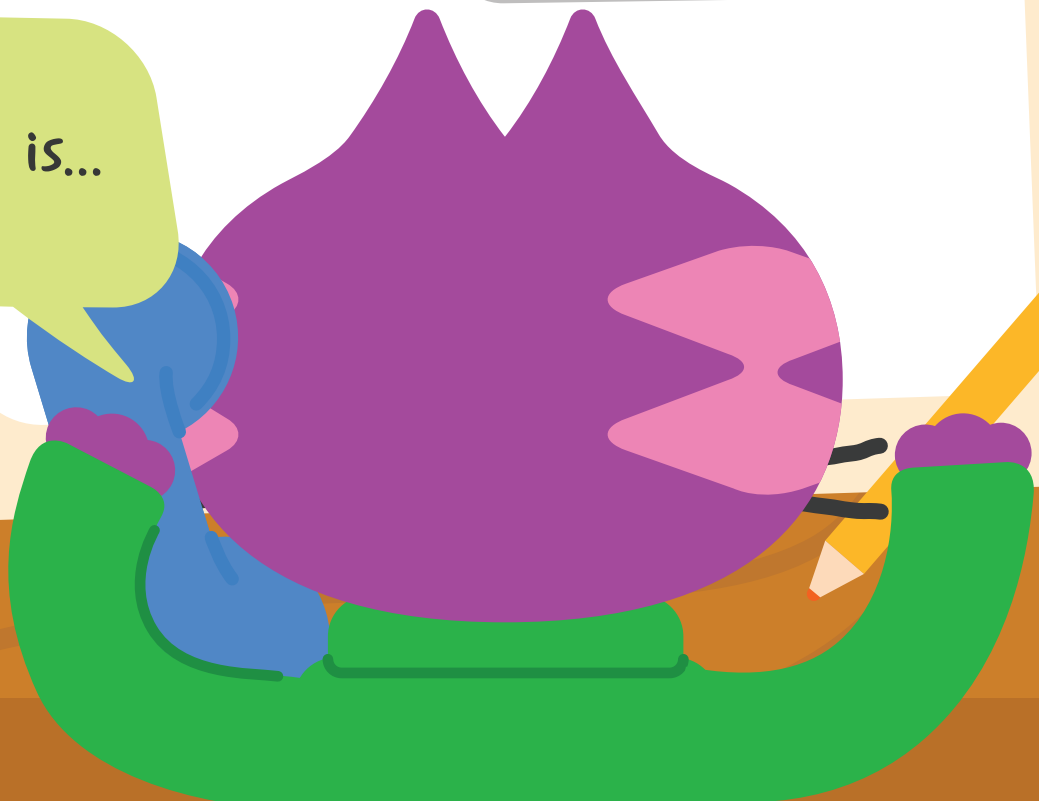
	1	2	3	4	5
Eyes					
Nose					
Hat					
Accessory					
Tail					

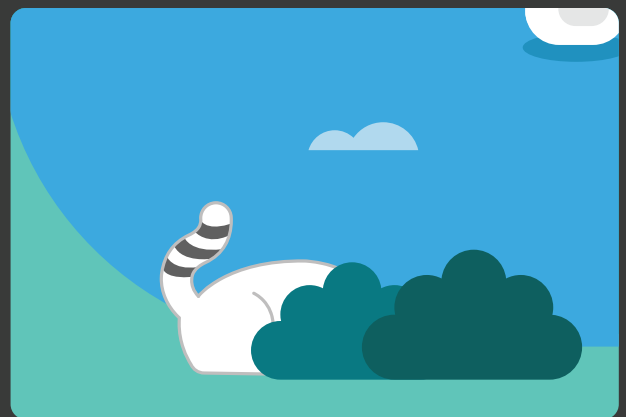
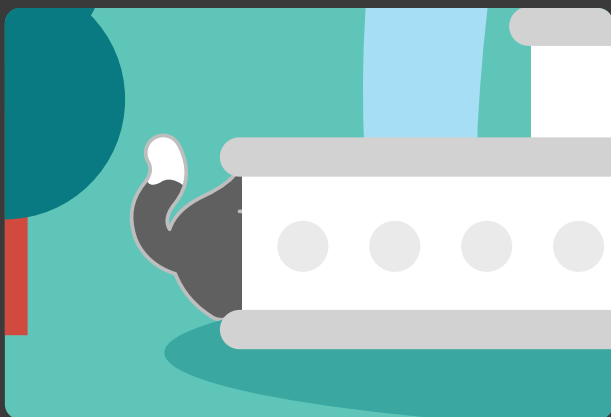
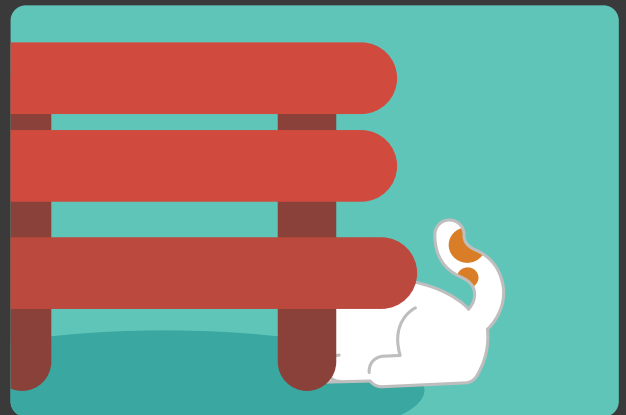
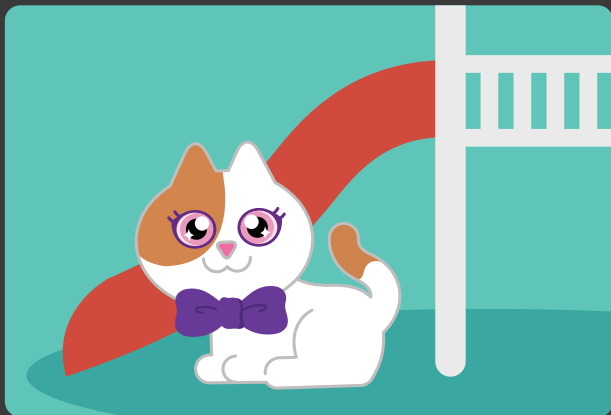
MISSING CAT

Lets' make a
cat montage!

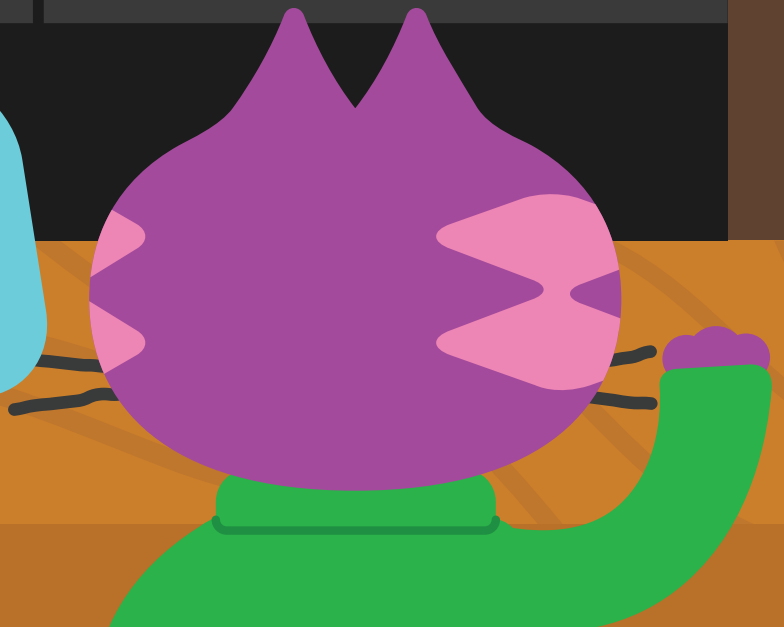


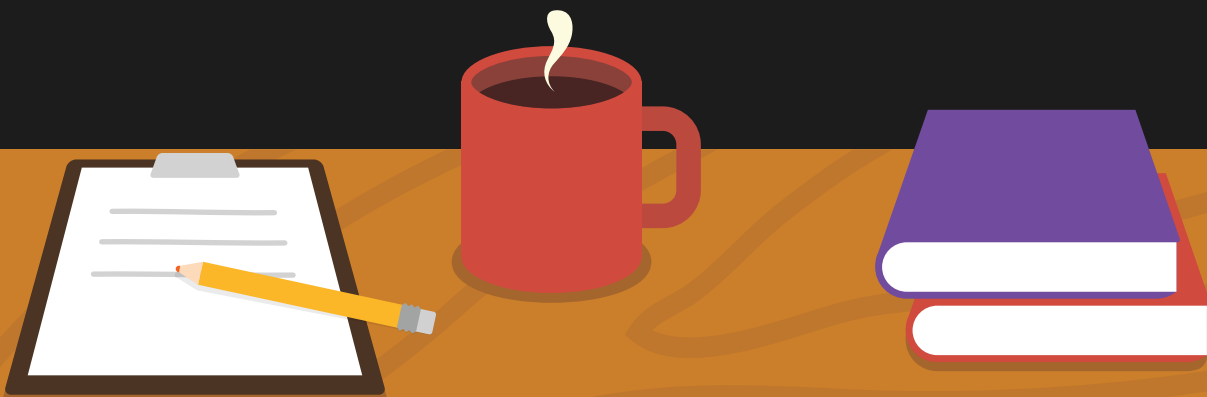
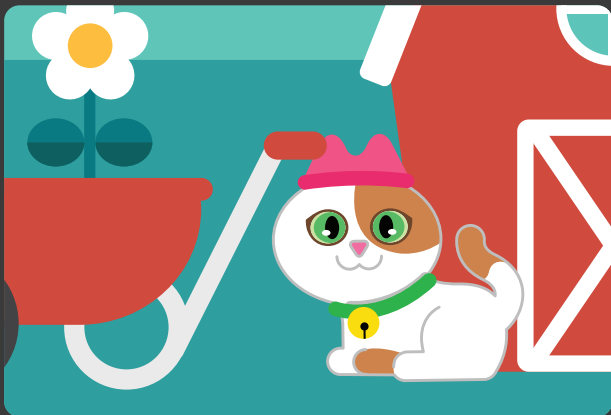
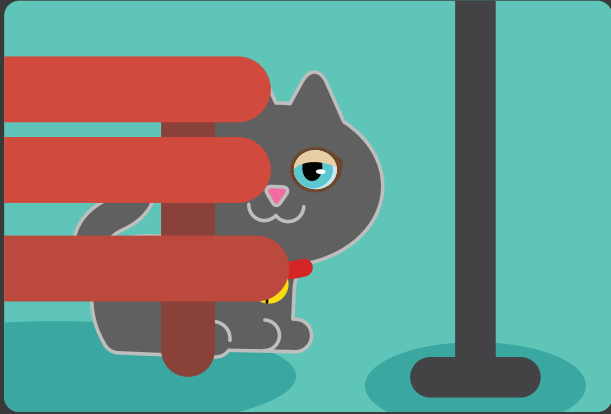
The cat is...





Let's find a cat
by watching
security cameras.









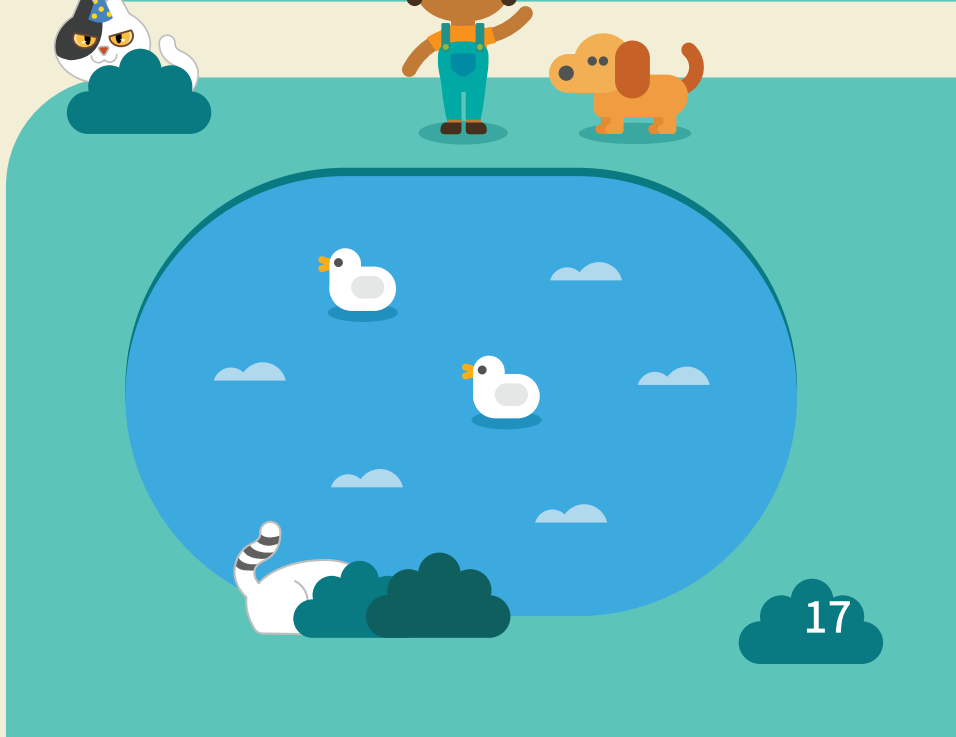
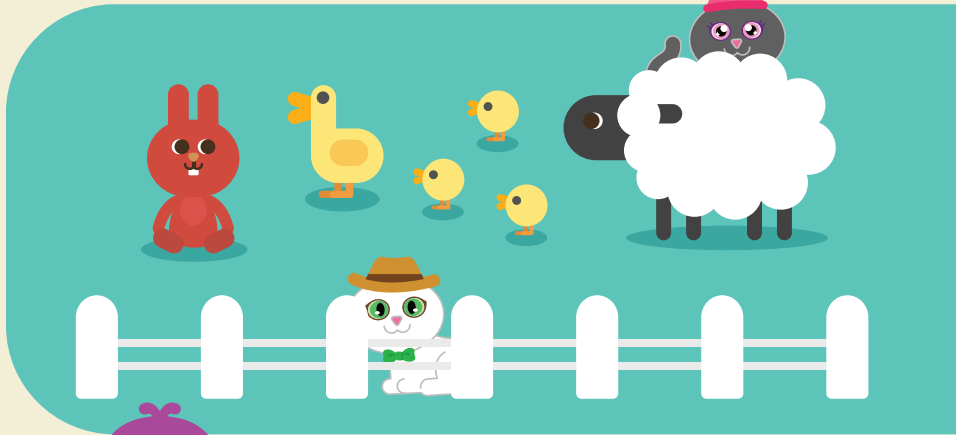
Complete the park by talking over
the phone with a witness.





Cats are hiding in the park. Please find cat Tom!





Witness 1

Yes, this is the detective office.

Someone took the cat to the bank.



Witness 2

Yes, this is the detective office.

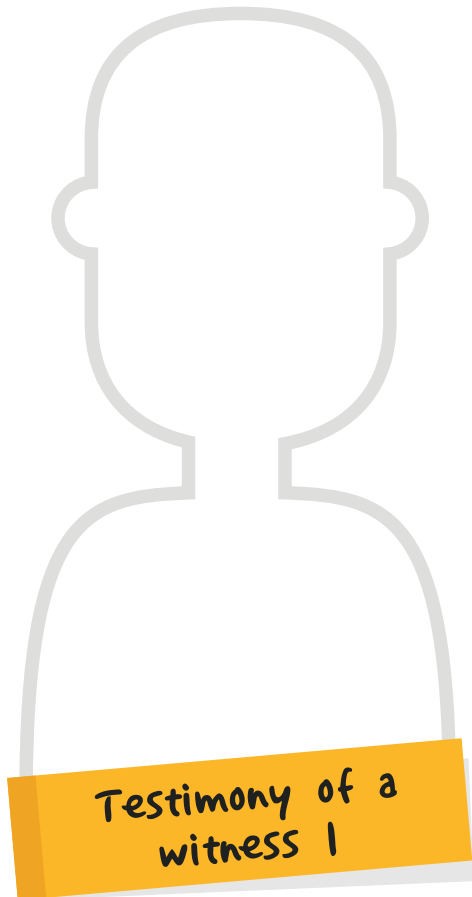
Someone was in a cafe with the cat.



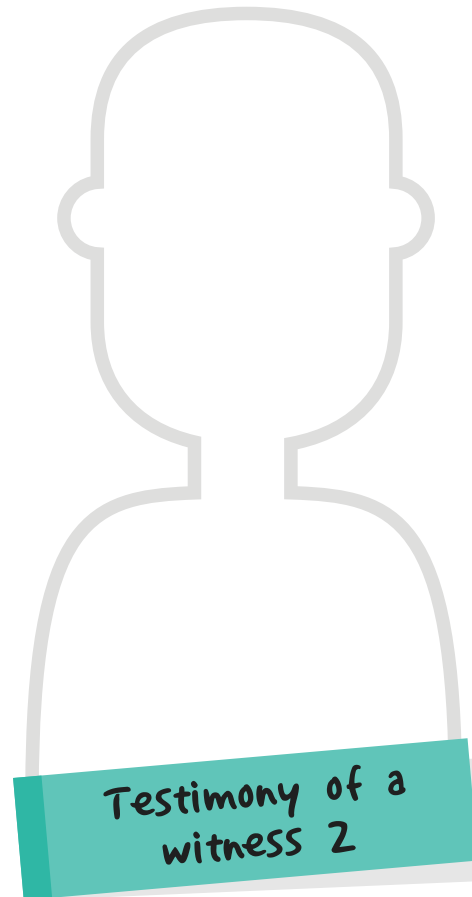
Who took the cat?
What should we ask the witnesses to find the cat?



"Let's listen testimony of witnesses and draw clothes of a criminal."



Around 3 o'clock, I saw someone
who entered the bank
holding a cat.
That person was wearing
yellow clothes.



Around 5 o'clock, I saw someone
grabbing a drink with the cat.
That person was wearing
purple clothes, hat, necklace,
and sunglasses. And that person
was holding a flower.



Find out places where the criminal went
with hints, heard from witnesses.

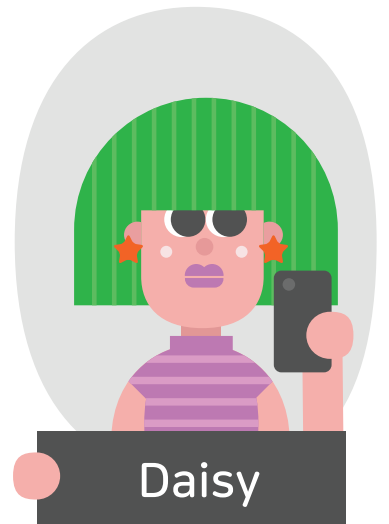
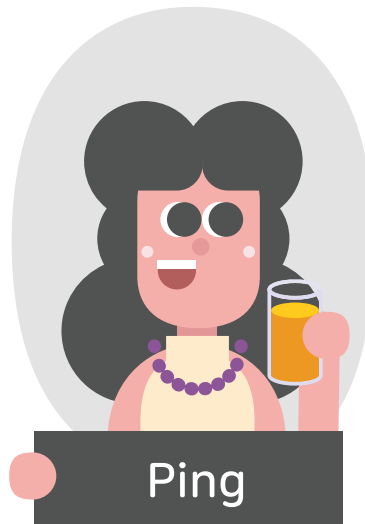
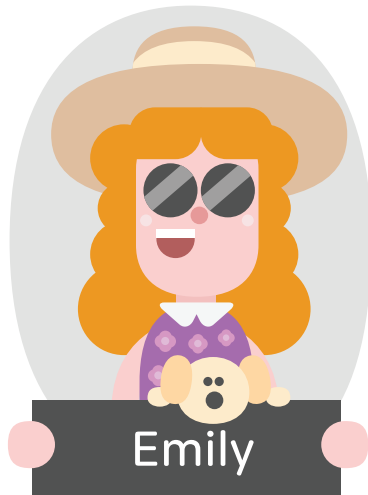
Draw arrows in sequence on stores where the criminal went.



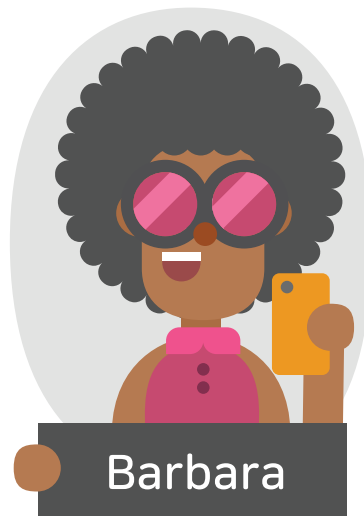
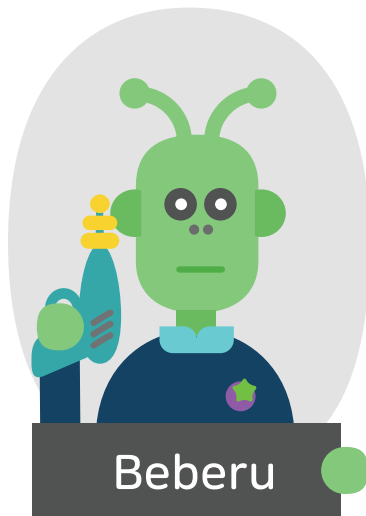
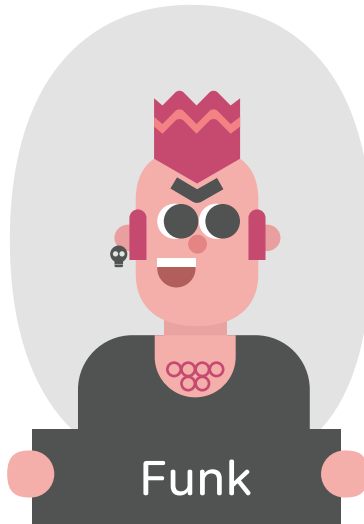




WANTED



\$1,000,000

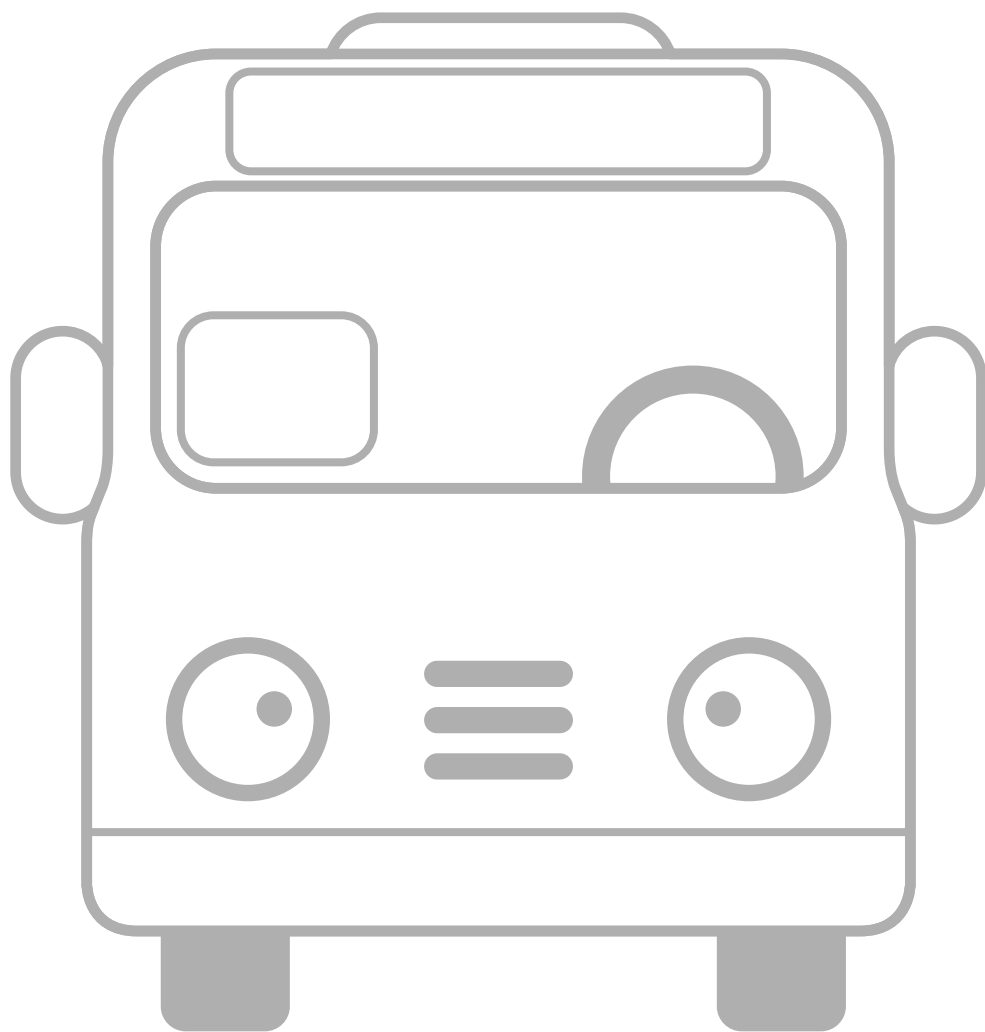


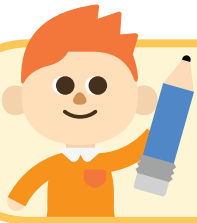


The person holding a cat
left the bus schedule.

★ Bus schedule ★

"Look at the bus schedule and draw color and pattern of the bus for right destination."





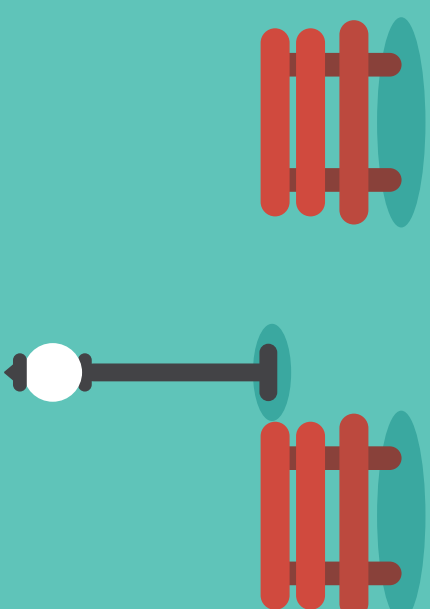
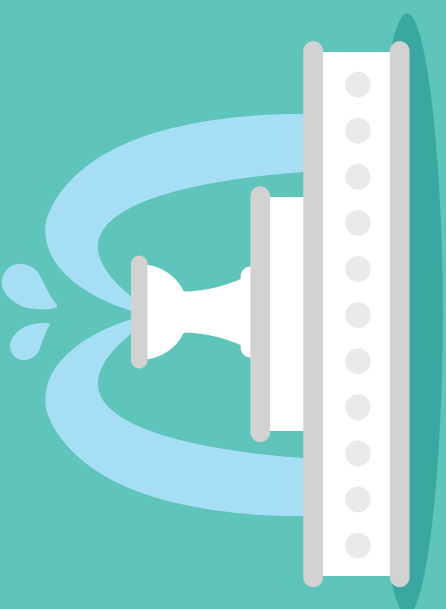
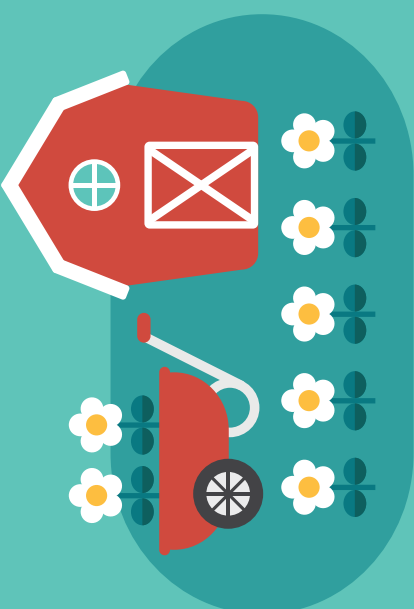
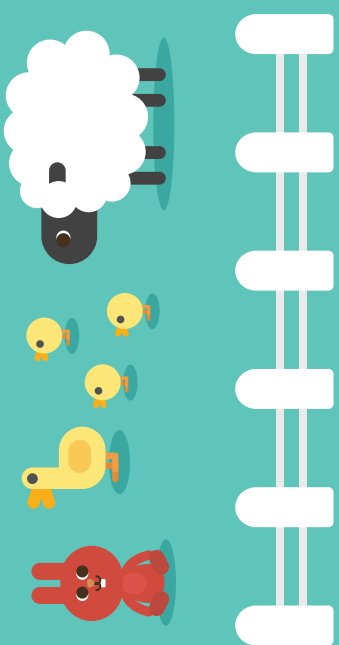
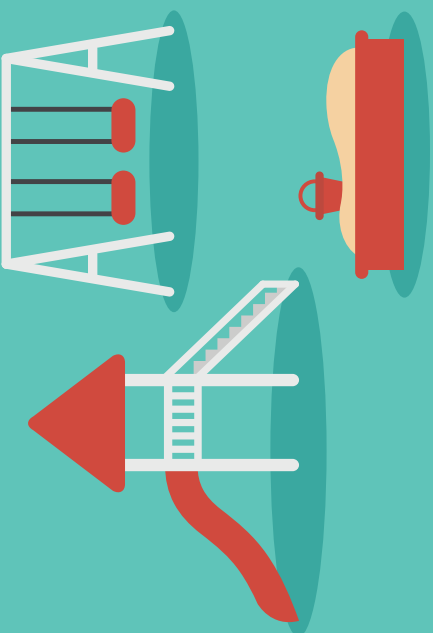
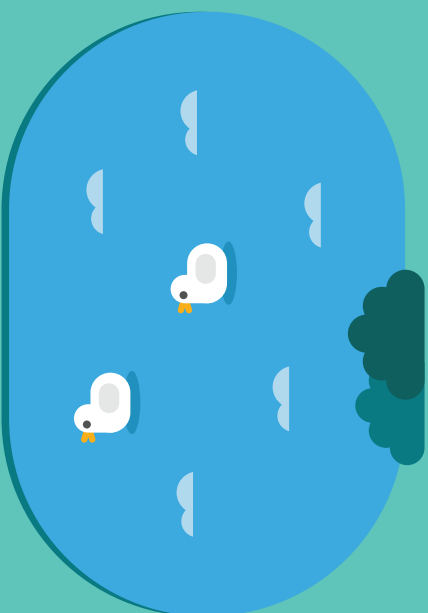
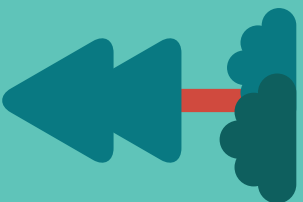
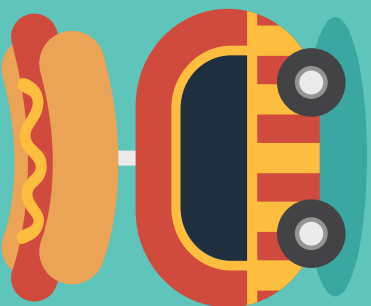
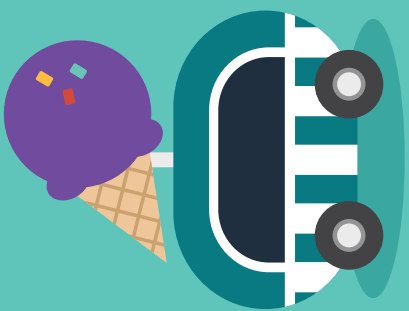
Draw 1 **red arrow** and 1 **blue arrow**
from one area to another.





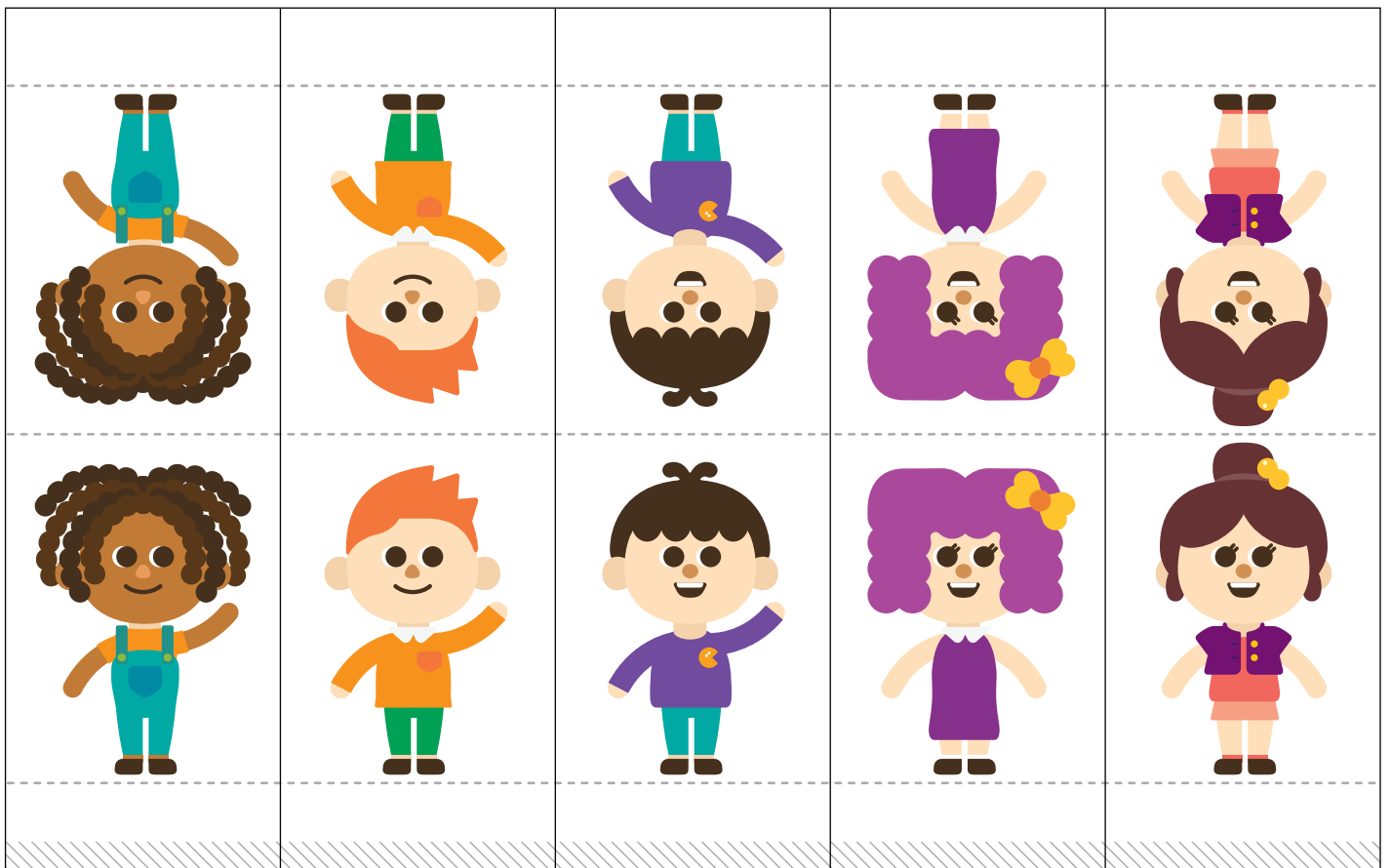
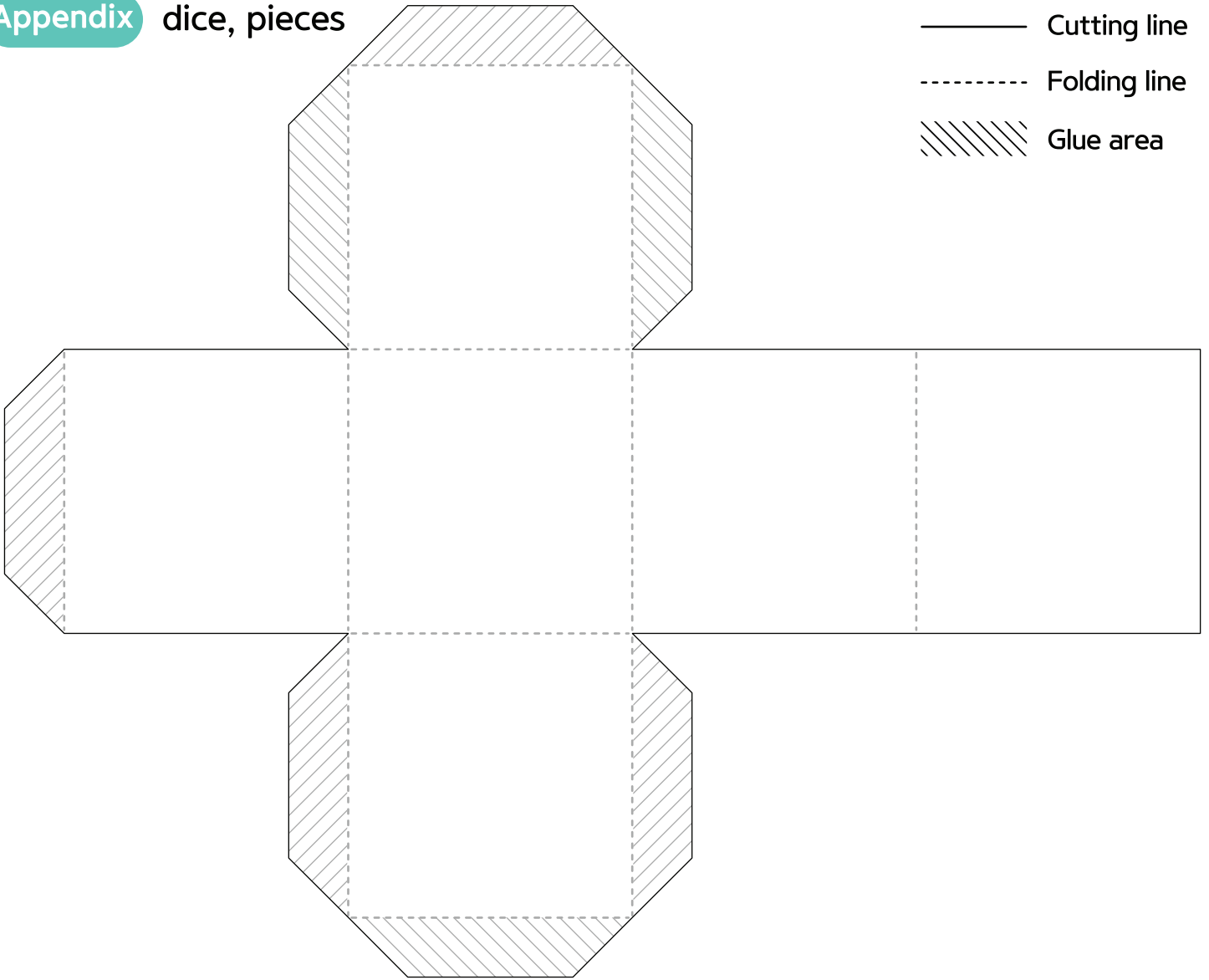
Thank you for
finding our Tom!





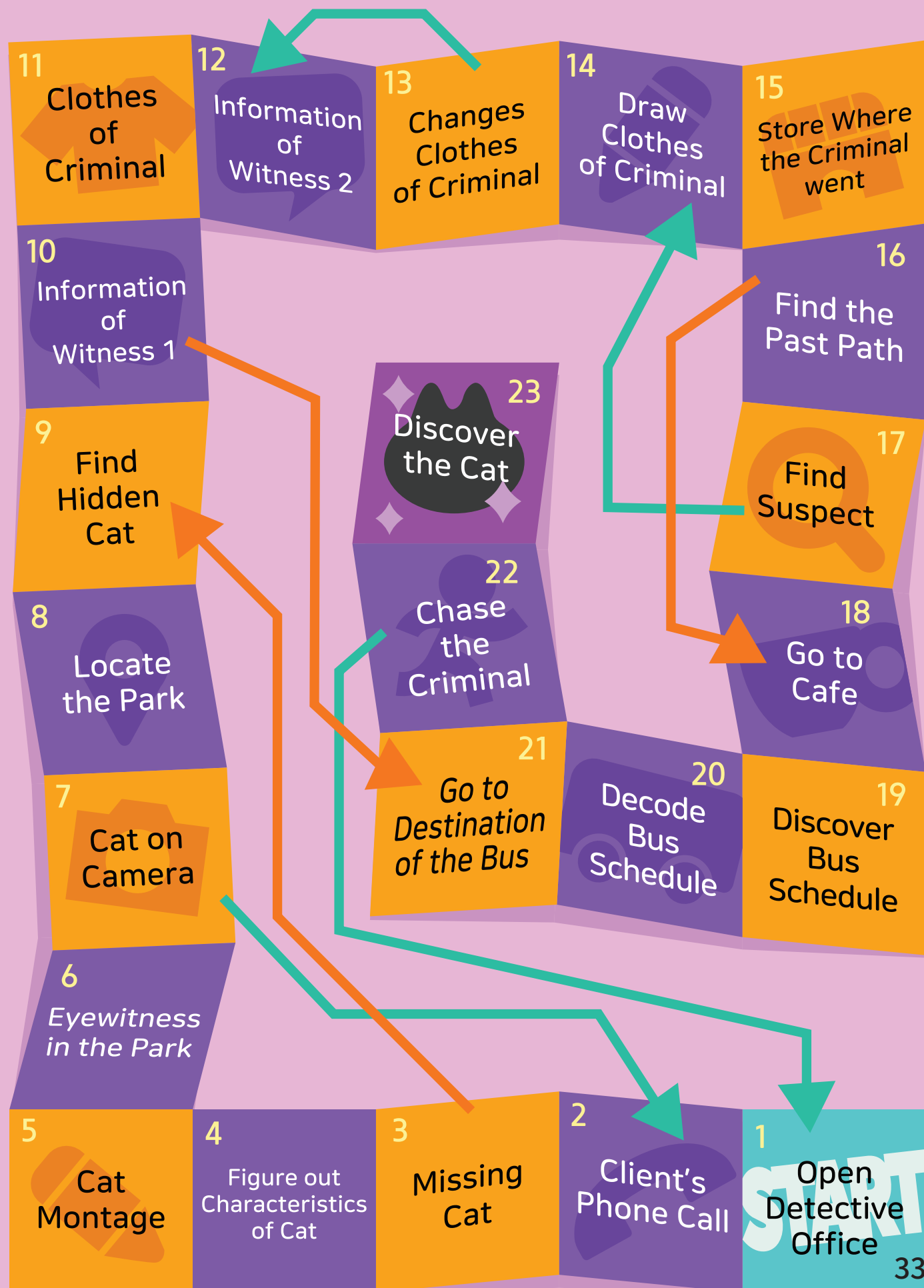
Appendix dice, pieces

- Cutting line
- Folding line
- /// Glue area



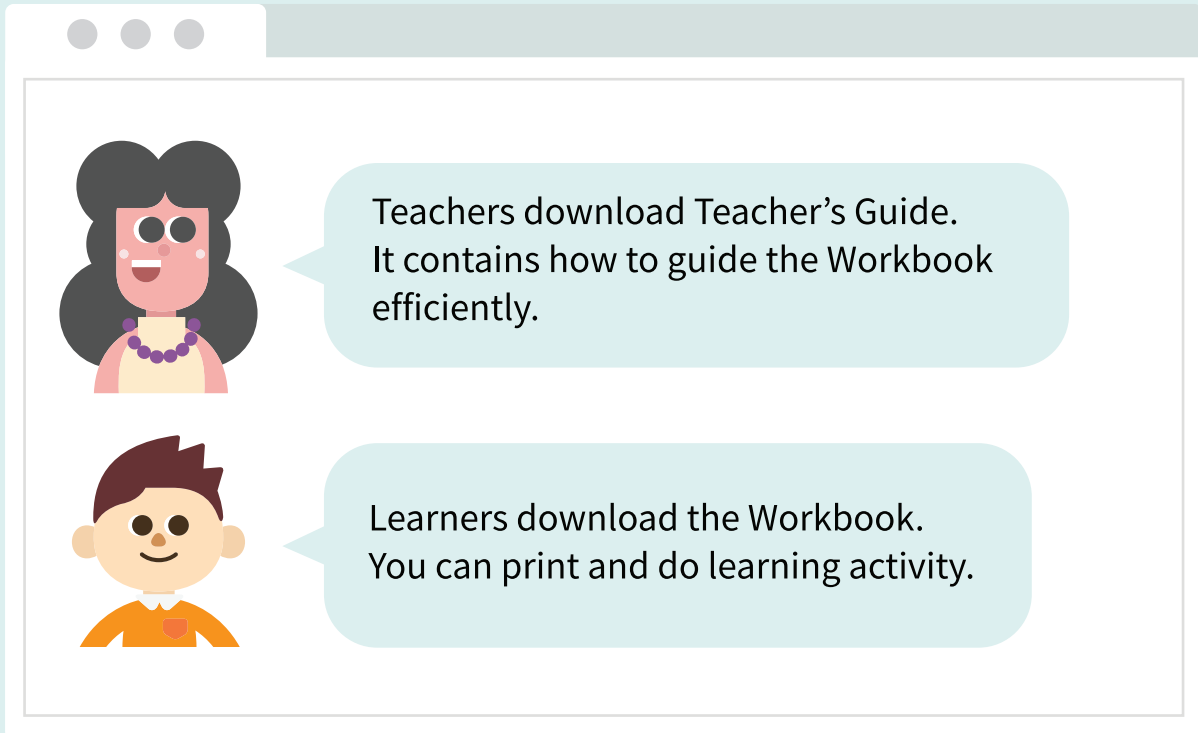
Let's find out that missing cat case was resolved in what order!

Roll the dice, move your game piece and follow arrow if there is. You win if you solve the case first!!



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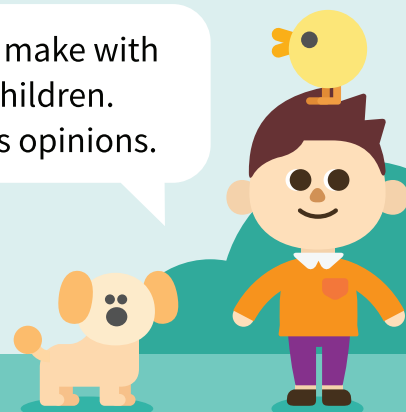
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Please give us various opinions.



Computational Thinking

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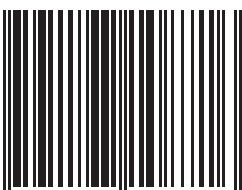
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